

SYSC 5108 Exam Study: Assignment 3, Question 5
Forward pass and backpropagation in a 1D CNN with max pooling
Source question: SYSC 5108 W26 Assignment 3.pdf, Question 5

Question Summary

The network is a one-dimensional convolutional neural network with:

- a shared ReLU filter applied at neurons 1, 2, 3, and 4,
- two max-pooling neurons 5 and 6, each pooling two neighboring activations with no overlap,
- one fully connected ReLU output neuron 7.

The input is

$$x = [1, 1, 1, 0, 0, 0].$$

The shared filter has bias and weights

$$w_0 = 0.75, \quad w_1 = 1, \quad w_2 = 1, \quad w_3 = -1.$$

The filter width is 3 and the stride is 1.

The final output neuron has

$$w_{7,0} = 0.1, \quad w_{7,5} = 0.2, \quad w_{7,6} = 0.5.$$

All neurons use ReLU:

$$\text{ReLU}(z) = \max(0, z).$$

The target output is

$$t = 1.$$

Step 1: Relate the Problem to the Course Material

Chapter 5 slides define convolutional neural networks using three core ideas:

- **Sparse interactions:** each neuron sees only a local window of the input.
- **Parameter sharing:** the same filter weights are reused across different locations.
- **Pooling:** nearby outputs are summarized, often by max pooling.

The slides also note that each kernel creates one feature map and that pooling produces a summary statistic of nearby outputs. That is exactly what happens in this problem.

Part (a): Compute the Network Output

1. Convolution windows

With filter width 3 and stride 1, the four windows are:

$$[1, 1, 1], \quad [1, 1, 0], \quad [1, 0, 0], \quad [0, 0, 0].$$

Each convolution neuron computes

$$z = w_0 + w_1x_1 + w_2x_2 + w_3x_3, \quad a = \text{ReLU}(z).$$

2. Convolution layer activations

Neuron 1 sees $[1, 1, 1]$:

$$z_1 = 0.75 + (1)(1) + (1)(1) + (-1)(1) = 1.75.$$

So

$$a_1 = \text{ReLU}(1.75) = 1.75.$$

Neuron 2 sees $[1, 1, 0]$:

$$z_2 = 0.75 + (1)(1) + (1)(1) + (-1)(0) = 2.75.$$

So

$$a_2 = 2.75.$$

Neuron 3 sees $[1, 0, 0]$:

$$z_3 = 0.75 + (1)(1) + (1)(0) + (-1)(0) = 1.75.$$

So

$$a_3 = 1.75.$$

Neuron 4 sees $[0, 0, 0]$:

$$z_4 = 0.75 + (1)(0) + (1)(0) + (-1)(0) = 0.75.$$

So

$$a_4 = 0.75.$$

Therefore the feature map is

$$[a_1, a_2, a_3, a_4] = [1.75, 2.75, 1.75, 0.75].$$

3. Max-pooling layer

Neuron 5 pools neurons 1 and 2:

$$a_5 = \max(1.75, 2.75) = 2.75.$$

Neuron 6 pools neurons 3 and 4:

$$a_6 = \max(1.75, 0.75) = 1.75.$$

4. Output neuron

Neuron 7 computes

$$z_7 = w_{7,0} + w_{7,5}a_5 + w_{7,6}a_6.$$

Substitute:

$$z_7 = 0.1 + (0.2)(2.75) + (0.5)(1.75).$$

So

$$z_7 = 0.1 + 0.55 + 0.875 = 1.525.$$

Apply ReLU:

$$a_7 = \text{ReLU}(1.525) = 1.525.$$

Therefore the network output is

$$\boxed{1.525.}$$

Part (b): Compute the Delta Values

Use squared error:

$$E = \frac{1}{2}(y - t)^2, \quad y = a_7.$$

Since

$$y = 1.525, \quad t = 1,$$

the output error signal is

$$\delta_7 = \frac{\partial E}{\partial z_7} = (y - t) \text{ReLU}'(z_7).$$

Because $z_7 = 1.525 > 0$,

$$\text{ReLU}'(z_7) = 1.$$

So

$$\delta_7 = (1.525 - 1)(1) = 0.525.$$

1. Backpropagate to pooling neurons

Neuron 5 connects to neuron 7 with weight $w_{7,5} = 0.2$, so

$$\delta_5 = w_{7,5}\delta_7 = (0.2)(0.525) = 0.105.$$

Neuron 6 connects to neuron 7 with weight $w_{7,6} = 0.5$, so

$$\delta_6 = w_{7,6}\delta_7 = (0.5)(0.525) = 0.2625.$$

2. Route max-pooling gradients

Max pooling sends the gradient only to the selected maximum.

Neuron 5 chose neuron 2 because

$$\max(a_1, a_2) = \max(1.75, 2.75) = a_2.$$

Therefore

$$\delta_2 = 0.105, \quad \delta_1 = 0.$$

Neuron 6 chose neuron 3 because

$$\max(a_3, a_4) = \max(1.75, 0.75) = a_3.$$

Therefore

$$\delta_3 = 0.2625, \quad \delta_4 = 0.$$

3. ReLU derivatives at convolution neurons

All convolution pre-activations are positive:

$$z_1 = 1.75, \quad z_2 = 2.75, \quad z_3 = 1.75, \quad z_4 = 0.75.$$

So every ReLU derivative is 1, and the deltas stay unchanged.

Hence the full set of deltas is

$\delta_1 = 0, \quad \delta_2 = 0.105, \quad \delta_3 = 0.2625, \quad \delta_4 = 0, \quad \delta_5 = 0.105, \quad \delta_6 = 0.2625, \quad \delta_7 = 0.525.$

Part (c): Calculate the Shared Filter Weight Updates

Because the filter is shared, the gradient for each filter parameter is the sum of its contribution from all four convolution neurons.

1. Bias term w_0

Each convolution neuron uses the same bias weight, so

$$\frac{\partial E}{\partial w_0} = \delta_1 + \delta_2 + \delta_3 + \delta_4.$$

Substitute:

$$\frac{\partial E}{\partial w_0} = 0 + 0.105 + 0.2625 + 0 = 0.3675.$$

2. Weight w_1

w_1 multiplies the first entry of each window. The first entries are

$$1, 1, 1, 0.$$

So

$$\frac{\partial E}{\partial w_1} = \delta_1(1) + \delta_2(1) + \delta_3(1) + \delta_4(0).$$

Thus

$$\frac{\partial E}{\partial w_1} = 0 + 0.105 + 0.2625 + 0 = 0.3675.$$

3. Weight w_2

w_2 multiplies the second entry of each window. Those entries are

$$1, 1, 0, 0.$$

So

$$\frac{\partial E}{\partial w_2} = \delta_1(1) + \delta_2(1) + \delta_3(0) + \delta_4(0).$$

Thus

$$\frac{\partial E}{\partial w_2} = 0 + 0.105 + 0 + 0 = 0.105.$$

4. Weight w_3

w_3 multiplies the third entry of each window. Those entries are

$$1, 0, 0, 0.$$

So

$$\frac{\partial E}{\partial w_3} = \delta_1(1) + \delta_2(0) + \delta_3(0) + \delta_4(0).$$

Thus

$$\frac{\partial E}{\partial w_3} = 0.$$

Therefore the shared-filter gradients are

$\frac{\partial E}{\partial w_0} = 0.3675,$	$\frac{\partial E}{\partial w_1} = 0.3675,$	$\frac{\partial E}{\partial w_2} = 0.105,$	$\frac{\partial E}{\partial w_3} = 0.$
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If the learning rate is α , then gradient descent gives

$$\Delta w_0 = -\alpha(0.3675), \quad \Delta w_1 = -\alpha(0.3675), \quad \Delta w_2 = -\alpha(0.105), \quad \Delta w_3 = 0.$$

Equivalently,

$$w_0^{\text{new}} = w_0 - \alpha(0.3675), \quad w_1^{\text{new}} = w_1 - \alpha(0.3675), \quad w_2^{\text{new}} = w_2 - \alpha(0.105), \quad w_3^{\text{new}} = w_3.$$

Why This Problem Matters

This question captures three CNN ideas that are easy to miss on an exam:

- **Convolution reuses one filter** at multiple positions, so gradients for shared weights must be added across all positions.
- **Max pooling is not linear.** During backpropagation, only the input that achieved the maximum receives the upstream gradient.
- **ReLU derivatives are simple.** If the pre-activation is positive, the derivative is 1; otherwise it is 0.

Concise Exam Answer

Convolution windows:

$$[1, 1, 1], \quad [1, 1, 0], \quad [1, 0, 0], \quad [0, 0, 0].$$

Convolution outputs:

$$z_1 = 1.75, \quad z_2 = 2.75, \quad z_3 = 1.75, \quad z_4 = 0.75,$$

so

$$a_1 = 1.75, \quad a_2 = 2.75, \quad a_3 = 1.75, \quad a_4 = 0.75.$$

Pooling gives

$$a_5 = \max(1.75, 2.75) = 2.75, \quad a_6 = \max(1.75, 0.75) = 1.75.$$

Output neuron:

$$z_7 = 0.1 + 0.2(2.75) + 0.5(1.75) = 1.525, \quad a_7 = 1.525.$$

Thus the output is

$$\boxed{1.525}.$$

With target $t = 1$,

$$\delta_7 = (1.525 - 1) = 0.525, \quad \delta_5 = 0.2(0.525) = 0.105, \quad \delta_6 = 0.5(0.525) = 0.2625.$$

Pooling routes these to the maxima:

$$\delta_2 = 0.105, \quad \delta_3 = 0.2625, \quad \delta_1 = \delta_4 = 0.$$

Therefore

$$\boxed{\delta_1 = 0, \delta_2 = 0.105, \delta_3 = 0.2625, \delta_4 = 0, \delta_5 = 0.105, \delta_6 = 0.2625, \delta_7 = 0.525}.$$

Shared-filter gradients:

$$\boxed{\frac{\partial E}{\partial w_0} = 0.3675, \quad \frac{\partial E}{\partial w_1} = 0.3675, \quad \frac{\partial E}{\partial w_2} = 0.105, \quad \frac{\partial E}{\partial w_3} = 0}.$$

Cross References Used

- Assignment: SYSC 5108 W26 Assignment 3.pdf, Question 5.
- Local submitted Assignment 3 PDF checked for comparison, Question 5 response.
- Slides: Chapter 5 slide deck on convolutional neural networks, convolution, stride, pooling, feature maps, and trainable parameters.
- Slides: Chapter 4 slide deck on ReLU activation functions.
- Textbook: Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 6 on feedforward networks and ReLU activations.
- Textbook: Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 9 on convolution, parameter sharing, and pooling concepts.